

EXP 5 Digital Signature with RSA

Server.java

```
import java.io.*;
import java.net.*;
import java.security.*;
import java.util.Base64;

public class Server {

    public static void main(String[] args) throws Exception {

        ServerSocket ss = new ServerSocket(5000);
        System.out.println("Server Started...");
        System.out.println("Waiting for client...");

        Socket s = ss.accept();
        System.out.println("Client Connected.");

        ObjectOutputStream out =
            new ObjectOutputStream(s.getOutputStream());

        BufferedReader br =
            new BufferedReader(new InputStreamReader(System.in));

        System.out.print("Enter Exam Paper Text: ");
        String examPaper = br.readLine();

        KeyPairGenerator keyGen =
            KeyPairGenerator.getInstance("RSA");

        keyGen.initialize(2048);

        KeyPair pair = keyGen.generateKeyPair();

        PrivateKey privateKey = pair.getPrivate();
        PublicKey publicKey = pair.getPublic();

        Signature signature =
            Signature.getInstance("SHA256withRSA");

        signature.initSign(privateKey);
        signature.update(examPaper.getBytes());

        byte[] digitalSignature = signature.sign();

        out.writeObject(examPaper);
        out.writeObject(digitalSignature);
        out.writeObject(publicKey);

        System.out.println("\nExam Paper Sent: " + examPaper);
        System.out.println("Digital Signature: " +
            Base64.getEncoder().encodeToString(digitalSignature));
    }
}
```

```

        System.out.println("Public Key Sent to Client.");

        out.close();
        s.close();
        ss.close();
    }
}

```

Client.java

```

import java.io.*;
import java.net.*;
import java.security.*;
import java.util.Base64;

public class Client {

    public static void main(String[] args) throws Exception {

        Socket s = new Socket("localhost", 5000);

        ObjectInputStream in =
            new ObjectInputStream(s.getInputStream());

        String examPaper = (String) in.readObject();

        byte[] digitalSignature =
            (byte[]) in.readObject();

        PublicKey publicKey =
            (PublicKey) in.readObject();

        Signature signature =
            Signature.getInstance("SHA256withRSA");

        signature.initVerify(publicKey);
        signature.update(examPaper.getBytes());

        boolean verified =
            signature.verify(digitalSignature);

        System.out.println("Received Exam Paper: " + examPaper);

        System.out.println("Received Signature: " +
            Base64.getEncoder().encodeToString(digitalSignature));

        if (verified) {
            System.out.println("\nSignature Verified Successfully");
            System.out.println("Exam Paper is Genuine and Not Modified");
        } else {
            System.out.println("\nSignature Verification Failed");
            System.out.println("Exam Paper is Modified or Fake");
        }
    }
}

```

```
}  
  
in.close();  
s.close();  
}  
}
```